

Physics-guided deep learning for rainfall-runoff modeling by considering extreme events and monotonic relationships

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ABSTRACT

Deep learning methods have recently shown a broad application prospect in rainfall-runoff modeling. However, the lack of physical mechanism becomes a major limitation in using machine learning methods that rely on the available labeled observations. To address this issue, the study proposes that synthetic samples are added to train the deep learning network by using three previously undiscussed physical mechanisms as follows: (1) extreme heavy rainfalls when the soil water is saturated, (2) long-duration rainless events when soil water is exhausted, and (3) the monotonic relationship between rainfall and runoff. A physics-guided Long Short-Term Memory (LSTM) network, shortly named PHY-LSTM, is then formulated. PHY-LSTM network is trained on 531 basins of the Catchment Attributes and Meteorology for Large-sample Studies (CAMELS) dataset, indicating that the performance is significantly improved compared to conventional LSTM. Specifically, the mean Nash-Sutcliffe Efficiency (NSE) improves from 0.52 to 0.61 from the daily simulations during the testing period in local models. It is demonstrated that synthetic samples can effectively improve the simulation of flood peaks and reduce the number of negative streamflow, and strong monotonicity is still maintained even if a slight disturbance is involved in the training dataset. The proposed PHY-LSTM shows that physical mechanisms are very useful to improve efficiencies of the data-driven rainfall-runoff model.

1. Introduction

The rainfall-runoff model has always been one of key researches in hydrology, which provides an important support for water resources (Guo et al., 2018; Li et al., 2020; Tajiki et al., 2020; Xie et al., 2020a), especially the prediction of flood (Gao et al., 2019; Hostache et al., 2018; Zhang et al., 2018) and drought disasters (Kang and Sridhar, 2017; Pan et al., 2020). The conventional rainfall-runoff model estimates streamflow based on hydrophysical processes (Chorley, 2019; Xie et al., 2020b). These sub-processes are driven by physical mechanisms in a nonlinear manner, including evapotranspiration, interception, infiltration, soil moisture and groundwater exchange, as well as the impact of water conservancy projects and other human activities (Sophocleous, 2002). However, the physical processes are so complex in hydrological models that it is difficult to discover the information from the available

inputs.

Deep learning has successfully overcome the once intractable problems driven by the increasingly powerful performance of computers and the availability of larger datasets (Xu et al., 2021), especially in computer vision (Nixon and Aguado, 2019; Weiss et al., 2020) and natural language processing (Chowdhary, 2020; Shen et al., 2020; Young et al., 2018). For example, Moghaddam et al. (2021) investigated the ability of diffuse structures in deep learning to infer effective hydraulic conductivity. Ehsani et al. (2021a) used radar information by machine learning and found that the Advanced Very High Resolution Radiometer (AVHRR) product can contribute to precipitation estimates at high latitudes. They further predicted dynamical spatiotemporal precipitation by using Recurrent Nowcasting-Net (RNC) and Convolutional Nowcasting-Net (CNC) architecture (Ehsani et al., 2021b). Simulations based on data-driven models have become a new solution for rainfall-

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runoff models (Shen, 2018; Tikhmarine et al., 2020; Yang et al., 2019) with the development of deep learning. Recent advances have been widely used in practice such as a Recurrent Neural Network (RNN) and its variant Long Short-Term Memory (LSTM) network, which achieved enhancements to predict runoff from meteorological observations compared to the conventional physics-based models (Feng et al., 2020; Hu et al., 2018). For example, Kratzert et al. (2018) tested the LSTM model in the catchments of the freely available Catchment Attributes and Meteorology for Large-sample Studies (CAMELS) dataset compared to the existing hydrological models. Safari et al. (2020) implemented the regression in the Reproducing Kernel Hilbert Space (RRKHS), which is highly flexible to provide better forecasts as compared with other alternatives. Researchers also focused on the improvements for the standard LSTM architecture, e.g. entity-aware-LSTM (Kratzert et al., 2019) and sequence-to-sequence modeling (Xiang et al., 2020). Kao et al. (2020) proposed Long Short-Term Memory based Encoder-Decoder (LSTM-ED) model for multi-step-ahead flood forecasts and increased the interpretability of the model.

However, the lack of physical mechanism becomes a major limitation of deep learning which relies on the available labeled observations. Since most deep learning models can only capture the correlation between variables, they do not understand the causal relationships. Karpayne et al. (2017a) described that data-driven and physical-based models represent the two extremes of knowledge discovery. Physical-based models are well suited to represent processes for which are conceptually well understood. On the other hand, conventional data-driven models rely on the information contained in the data. Specifically, a deep learning model can fit the observed data well, but the prediction may not be physically consistent and then even a slight disturbance can lead to large changes (Liu et al., 2017; Reichstein et al., 2019). Therefore, physics-guided deep learning models are possible solutions to the problem at hand. It would be of special interest to explore the continuum between deep learning models and physical mechanisms, where both physics and data are used in a synergistic manner.

Recently, there has been an increasing interest in the achievement of physical consistency by training models about the rules of physical processes in water resources, which aims to improve the performance of deep learning models in realizing theoretical discovery (Karpayne et al., 2017a). One of the methods was the output of a physically hydrological model used as another input in the neural network model along with the drivers (Konapala et al., 2020; Yang et al., 2020). For example, Yang et al. (2020) developed an improved model in data-limited basins by combining artificial neural networks with synthetic runoff data in a distributed hydrological model. Another hybrid modeling strategy was used for correcting mismatch, where physical knowledge was implanted into the deep learning model (Farfán et al., 2020). Jiang et al. (2020) developed a hydrology-aware deep learning model by a P-RNN layer wrapped with physically meaningful parameters in the EXP-HYDRO model. Additionally, some studies used the physical constraints as loss terms in the artificial neural network, which has been successfully applied in the prediction of the lake water temperature and the sub-surface flow (Read et al., 2019; Wang et al., 2019). The hybrid modeling framework was used to predict the lake water temperature at different depths by integrating energy balance constraints (Karpayne et al., 2017b; Read et al., 2019). Physical mechanisms were also used to train the deep learning model to predict groundwater (Wang et al., 2019), where the neural network was guided by the basic theory, such as control equations, boundary conditions, other physical constraints, engineering control, and expert experience. Consequently, it has become an essential question that how to develop a quantitative understanding of the theoretical process to achieve physical consistency in a hydrological model increasingly dominated by deep learning (Nearing et al., 2020).

However, to the best of our knowledge, there is little attention paid to the extreme events and monotonic relationships of a deep learning

Table 1

Table of forcing inputs and static catchment attributes used in this study.

	Variable name	Description	Unit
Forcing inputs	PRCP	Precipitation	mm/day
	SRAD	Solar radiation	W/m ²
	Tmax	Maximum temperature	°C
	Tmin	Minimum temperature	°C
Static inputs	Vp	Vapor pressure	Pa
	P_mean	Mean daily precipitation	mm
	Pet_mean	Mean daily potential evapotranspiration	mm
	Aridity	Ratio of mean PET to mean precipitation	—
	P_seasonality	Seasonality and timing of precipitation.	mm
	Frac_snow_daily	Fraction of precipitation falling on days with temperatures below 0 °C	—
	High_prec_freq	Frequency of high-precipitation days (≥ 5 times mean daily precipitation)	—
	High_prec_dur	Average duration of high-precipitation events	day
	Low_prec_freq	Frequency of dry days ($< 1\text{mm d}^{-1}$).	—
	Low_prec_dur	Average duration of dry periods	day
	Elev_mean	Catchment mean elevation	m
	Slope_mean	Catchment mean slope	m/km
	Area_gages2	Catchment area	km ²
	Frac_forest	Forest fraction	—
	Lai_max	Maximum monthly mean of leaf area index	—
	Lai_diff	Difference between the max. and min. mean of the leaf area index	—
	Dom_land_cover_frac	Fraction of the catchment area associated with the dominant land cover	—
	Dom_land_cover	Dominant land cover type	—
	Soil_depth_pelletier	Depth to bedrock (maximum 50 m)	m
	Soil_depth_statgso	Soil depth (maximum 1.5 m)	m
	Soil_porosity	Volumetric porosity	—
	Soil_conductivity	Saturated hydraulic conductivity	cm/hr
	Max_water_content	Maximum water content of the soil	m
	Sand_frac	Fraction of sand in the soil	—
	Silt_frac	Fraction of silt in the soil	—
	Clay_frac	Fraction of clay in the soil	—
	Carb_rocks_frac	Fraction of Carbonate sedimentary rocks	—
	Geol_permeability	Surface permeability (log10)	m ²

network for rainfall-runoff simulation. Three physical mechanisms are therefore considered to constrain the deep learning network from high flow, low flow, and monotonic properties, which is the extreme events and internal relationships that are rare in historical observations but are obvious based on physical mechanisms. Specifically, we address the following objectives: (1) How to educate a deep learning network the extreme events of a heavy rainstorm; (2) How to educate a deep learning network the extreme events of long-duration rainless; (3) How to improve the simulation by considering the monotonic relationship between rainfall and runoff.

The remainder of this paper is organized as follows: Details on the chosen catchments are described in Section 2. Section 3 describes the methodology for the data-driven model based on physical mechanisms, followed by results and discussion in Section 4. Lastly, conclusions are drawn in Section 5.

2. Data

The basins used in this study were obtained from the CAMELS dataset in the continental United States, which was an international

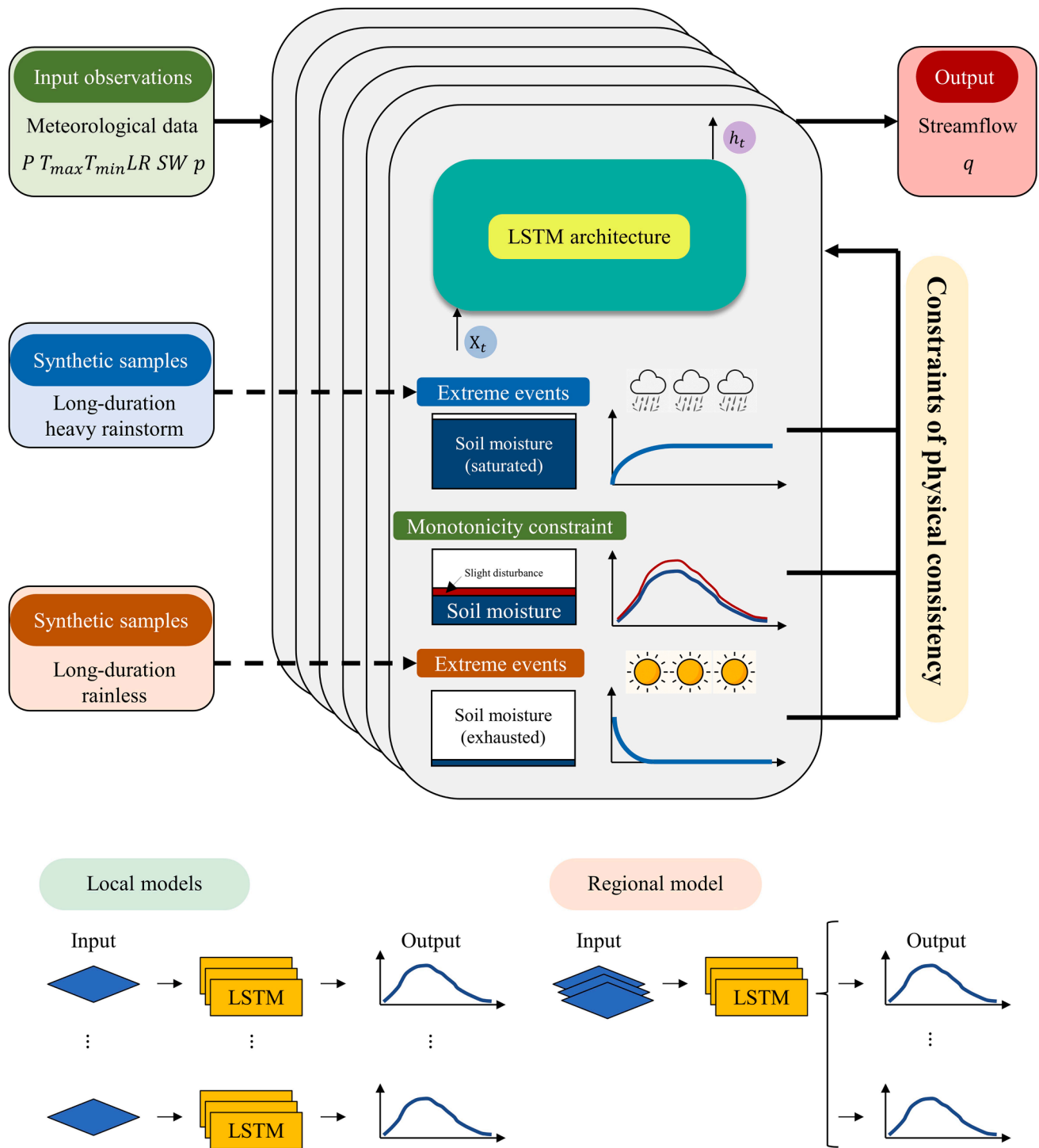


Fig 1. The flowchart for the rainfall-runoff model using a physics-guided deep learning network.

cooperation project for hydrologic and atmospheric models (Addor et al., 2017; Newman et al., 2015). The dataset, which consisted of the daily meteorological forcing inputs, the catchment attributes, and streamflow observations, was available for the comparative study of large samples. The CAMELS contained meteorological inputs of four levels including 18 hydrologic units, 671 basins, more detailed Hydrologic Response Unit (HRU) regions, and different elevation bands. The CAMELS basins range in size between 4 and 25000 km² and they have relatively low anthropogenic impacts.

In this study, the selection of 531 basins was selected from the CAMELS dataset, which was the same as Newman et al. (2017). These

basins were selected out of the full set based on the following criteria: (1) Areas of the basins were smaller than 2000 km²; (2) There was a discrepancy less than 10% between different methods of calculating basin area.

The models were trained with an updated version of the Maurer forcing data (Kratzert et al., 2019; Maurer et al., 2002). The daily meteorological input data were available including precipitation, maximum temperature, minimum temperature, average short-wave radiation, and vapor pressure. The streamflow at the outlet was measurements from the USGS gauging stations. Additionally, as Table 1 shows, the 27 catchment attributes were concatenated as static inputs to

the meteorological forcings at each time step. Data division was the same as previous studies (Kratzert et al., 2019) for a fair comparison. The training dataset lasted from 1 October 1999 through 30 September 2008, and the testing dataset lasted from 1 October 1989 through 30 September 1999. Data normalization is generally performed during the data preprocessing step. In this study, min–max normalization is used as the rescaling method.

3. Methodology

Fig. 1 provides an overview of the proposed physics-guided deep learning network, which develops the methodology as follows:

- (1) Three physical mechanisms were used to increase the physical interpretability of the deep learning networks by adding synthetic samples (Section 3.1).
- (2) The physical constraints were implemented into deep learning networks of local and regional scales by the exterior terms imposed on the loss function (Section 3.2).
- (3) Simulation accuracy and physical consistency were evaluated by using varied statistical criteria (Section 3.3).

3.1. Physical mechanisms for rainfall-runoff model

3.1.1. Extreme events of the long-duration heavy rainstorm

To establish physical mechanisms for rainfall-runoff modeling, the law of conservation of water is used. The water balance equation controls the hydrological cycle by describing the flow into and out of the system. The runoff yield from a given rainstorm during the time t can be calculated as follows:

$$q_t = p_t - e_t - \Delta s_t \quad (1)$$

where the amount of water that a watershed receives is the precipitation P_t [mm] and the amount of water consumption has three parts: evapotranspiration e_t [mm], an increase in soil moisture content Δs_t [mm], and the runoff q_t [mm]. The soil moisture increases from the initial soil water content s_0 to the soil water content s_t at the end of rainfall. Therefore, it can be described as follows:

$$q_t = p_t - e_t \approx p_t \text{ as } p_t \rightarrow p^\infty, s_t = s_m. \quad (2)$$

where p^∞ are the long-duration heavy rainstorm, and the soil wetting reaches its maximum saturated states s_m . The runoff is equal to the heavy rainfall which is much greater than the ignored evapotranspiration when soil moisture has been saturated.

However, the probability of heavy rainfall occurrence is very low in the historical data, which leads to a large error in the simulation of the peak flow when using deep learning networks. To solve this problem, synthetic samples, also called artificial samples, are constructed to improve learning performance. Synthetic samples, different from the actual samples, can provide special information about the physical events and relationships. The data pairs for the synthetic samples are the rainfall and the runoff determined by physical mechanisms.

To train the deep learning model to deal with the extreme events of the long-duration heavy rainstorm, the made-up year of the synthetic samples is added into the actual observation series. Combinations of synthetic samples are considered with varied rainfall intensities and varied durations. The labels are its streamflow when all rainfall yields runoff. Therefore, the function for heavy storms is used by the following equations.

$$\hat{q}_t = f[X_t, \tau] = f[X_{t-\tau}, X_{t-\tau+1}, \dots, X_{t-1}, X_t] \quad (3)$$

$$f[p_{t,i}^\infty, \tau] - p_{t,i}^\infty = 0 \quad t > t_F. \quad (4)$$

$$MSE_\infty = \frac{1}{NT_u N_\infty} \sum_{t=1}^{NT_u} \sum_{i=1}^{N_\infty} \{f[p_{t,i}^\infty, \tau] - p_{t,i}^\infty\}^2. \quad (5)$$

where $\hat{q}_t$ donates the simulated streamflow; f is the simulation function of the deep learning method; X_t is the normalized inputs; τ is the input sequence length, and models are run in the sequence-to-one mode so that to simulate a daily discharge required meteorological forcings from τ preceding days.; $p_{t,i}^\infty$ is the rainstorm events of different intensity; t_F is the time when soil water is saturated; MSE_∞ is the Mean Square Error (MSE) for heavy rainstorm events; NT_u is the duration of heavy rainstorm samples; and N_∞ is the number of varied rainstorm intensities.

3.1.2. Extreme events of the long-duration rainless

The streamflow during long-duration rainless events determines the lower boundary condition for rainfall-runoff modeling. There is a relatively rapid hydrograph rise to a peak, which usually occurs at or soon after the input ceases, followed by a hydrograph recession. A recession period lasts until streamflow begins to increase again due to subsequent rainfall. If the rainless continues long enough, streamflow only includes baseflow (Cheng et al., 2020). Therefore, the lower boundary condition of the rainfall-runoff model is that the base flow will be exhausted and the basin will not yield runoff after the extreme long-duration rainless events.

The flow is never negative in the physical world. There are often negative values when using deep learning networks, which usually can be solved by setting the linear activation function of the output layer to Rectified Linear Unit (ReLU). This physics-guided LSTM model attempts to track the water balance of the baseflow in the recession period. However, the magnitude of baseflow varies in each different watershed, and it is difficult to be determined as a constant. Therefore, when an extreme hypothetical case of prolonged absence of rainfall is considered, the baseflow is completely depleted to zero. Therefore, to train the physics-guided deep learning model to prevent negative values, synthetic samples of long-duration rainless events are added to the training dataset. The soil moisture is exhausted and the corresponding runoff is equal to zero when the basin experiences a long-duration rainless situation. The synthetic samples are long-duration rainless and its streamflow when the soil moisture dries up. The function for rainless events is used by the following equations:

$$f[p_t^0, \tau] = 0 \text{ as } p_t^0 = 0, \quad t \rightarrow \infty \quad (6)$$

$$MSE_0 = \frac{1}{NT_l} \sum_{t=1}^{NT_l} \{f[p_t^0, \tau]\}^2 \quad (7)$$

where p_t^0 is the precipitation of long-duration rainless events (i.e., 0 mm); MSE_0 is the mean square error in response to the rainless events; and NT_l is the sample length of long-duration rainless events.

3.1.3. Monotonic relationship between rainfall and runoff

The internal relationship between rainfall and runoff is considered to ensure that the physical laws are upheld in the runoff simulation of a deep learning model. The rainfall-runoff relationship for any rainstorm depends on the dynamic interaction between rain intensity, evapotranspiration, infiltration, and storage. The relationship between rainfall and runoff should be presented monotonously, i.e., the runoff should be larger if there was a small increase in the rainfall at the current time step. However, the simulation cannot guarantee the monotonicity of simulation by conventional deep learning models, which is physically inconsistent with the water balance equation. Prediction errors can be very large even if a slight disturbance to the training dataset in terms of the model without physical consistency. Essentially, this is because the deep learning model ignores the complex physical processes and underlying surface conditions when it is trained as a black box for fitting the highly nonlinear system. It can only capture the correlation between

Table 2

Hyperparameters for LSTM, weights, and parameters for synthetic samples.

Hyperparameter		Local model	Regional model
Hyperparameter for LSTM	Epoch	200	30
	Hidden size	40	256
	Hidden layer	2	1
	Dropout rate	0.4	0.4
	Batch size	1/10 of the total	1/12 of the total
	Normalization	Min-max	Min-max
Weight	Lag time (days)	20	270
	λ_{data}	0.6	0.6
	λ_{∞}	0.1	0.1
	λ_0	0.1	0.1
	λ_{mcp}	0.1	0.1
	λ_{mce}	0.1	0.1
Parameter for rainstorm samples	Intensity (mm)	50–150	50–150
	Duration (days)	10–100	10–100

variables without physical mechanisms, therefore it cannot understand the cause and effect. Furthermore, the function of the monotonic relationship between rainfall and runoff is as follows:

$$f[p_{t,i} + \Delta p_i, \tau] - f[p_{t,i}, \tau] \geq 0. \quad (8)$$

$$MSE_{mcp} = \frac{1}{NT_{mcp} N_{mcp}} \sum_{t=1}^{NT_{mcp}} \sum_{i=1}^{N_{mcp}} \{ReLU\{f[p_{t,i}, \tau] - f[p_{t,i} + \Delta p_i, \tau]\}\}^2 \quad (9)$$

where Δp_i is a small increase in rainfall; MSE_{mcp} is the mean square error for the monotonic relationship; NT_{mcp} is the sample length of the disturbed rainfall; N_{mcp} is the number of different levels of the increase (from small to large); and $ReLU$ is the response function, excluding the monotonic term and retaining the nonmonotonic term. Besides, the relationship between evapotranspiration and runoff is also monotonic as follows:

$$f[e_{t,i} + \Delta e_i, \tau] - f[e_{t,i}, \tau] \leq 0. \quad (10)$$

$$MSE_{mce} = \frac{1}{NT_{mce} N_{mce}} \sum_{t=1}^{NT_{mce}} \sum_{i=1}^{N_{mce}} \{ReLU\{f[e_{t,i}, \tau] - f[e_{t,i} + \Delta e_i, \tau]\}\}^2 \quad (11)$$

where Δe_i is a small increase of evapotranspiration; MSE_{mce} is the mean square error for the monotonic relationship of the evapotranspiration, NT_{mce} is the sample length of the disturbed evapotranspiration; and N_{mce} is the number of different levels of the increase.

3.2. Deep learning network based on physical consistency

Sufficient sample sizes are very important to construct effective deep learning models, but heavy rainstorms and long-duration rainless samples are rare in the historical data. To overcome this challenge, synthetic samples that follow three physical mechanisms are used as the additional input to train the model. In this study, physical consistency is considered to the deep learning model by using domain-oriented constraints inspired by three physical understandings. All model configurations are trained using the following MSE as loss function. The exterior loss function of the hydrological physical process is added to the conventional objective of the model by the following equations:

$$Loss = \lambda_{data} MSE_{data} + \lambda_{\infty} MSE_{\infty} + \lambda_0 MSE_0 + \lambda_{mcp} MSE_{mcp} + \lambda_{mce} MSE_{mce} \quad (12)$$

$$MSE_{data} = \frac{1}{NT} \sum_{t=1}^{NT} \{\hat{q}_t - q_t\}^2 \quad (13)$$

where $Loss$ is the loss function; λ_{data} is the weight coefficient for simulated runoff; MSE_{data} is the mean square error of simulated runoff by deep learning method; λ_{∞} is the weight coefficient for heavy rainstorm events; λ_0 is the weight coefficient in response to long-duration rainless events; MSE_{mcp} and MSE_{mce} are the loss functions for the monotonicity including rainfall and evapotranspiration; λ_{mcp} and λ_{mce} are the corresponding weight coefficients for the monotonicity; and NT is the length of time series. The final configuration is chosen by taking the weights that resulted in the highest mean NSE. As shown in Table 2, the initial proportion of physical mechanisms have the same value, i.e., the four constraints are treated equally. The final weights are set to {0.6, 0.1, 0.1, 0.1, 0.1} for the five MSEs in this study. It should be noted that the final weights are the result of the optimal NSE by assessing the relative importance of different weights.

LSTM networks are designed to avoid the weakness of the conventional RNN to learn long-duration dependencies (Cheng et al., 2016; Hochreiter and Schmidhuber, 1997). The deep learning networks are developed based on open-source software. The programming language of choice is Python 3.7. PyTorch framework and Nvidia Tesla V100 16 GB GPUs are used for training.

3.2.1. Experimental 1: Local models

LSTM networks are applied independently to each watershed in local models. Hyperparameters and sample parameters are needed to be determined in establishing the proposed physics-guided LSTM networks. The hyperparameters will influence the structure and training of LSTM networks. For all local LSTMs and PHY-LSTMs, the same architecture is used. Different settings and models are trained and tested where the optimal hyperparameter is found. In local models, the number of hidden layers is set to 2, the hidden size is set to 40, and the number of the batch is 1/10 of the number of samples. The dropout rate is set at 0.4, and the training epochs are 200. The input sequence length τ is set to 20 preceding days. Besides, the parameter of samples will influence the implementation of physical mechanisms. Five different intensity settings and 10 different duration settings are added for a total of 50 different synthetic samples where the total MSE is the mean of the 50 scenarios. The long-duration heavy rainstorm events^{sp} have different rainstorm intensities of 50 mm, 75 mm, 100 mm, 125 mm, and 150 mm, respectively. The duration of the made-up period of the synthetic samples is 10–100 days in each batch of training samples, which are equally divided every 10 days.

3.2.2. Experimental 2: A regional model

The regional model extends the simulation ability across the similarity behaving catchments in large regions (Hestness et al., 2017; Kratzert et al., 2018). Huge training data of 531 basins in the CAMELS are used to learn more general rainfall-runoff relationships only in an LSTM network, which is the potential for predictions in the ungauged basins. The same architecture is used for all regional LSTMs and PHY-LSTMs. The structure of the regional model is different from that of the local model since the scale of the input data to the model differs significantly. The hyperparameters optimization can refer to the previous research in Kratzert et al. (2019). In the regional model, the number of hidden layers is set to 1, the number of the batch is 1/12 of the number of samples, and the training epochs are 30. The LSTMs had 256 memory cells and a single fully-connected layer with a dropout rate of 0.4. The LSTMs were run in sequence-to-value mode with the input sequence length τ is set to 270 preceding days.

Table 3

Comparison of NSE statistics for the testing dataset derived from different models, either developed by this study or referred to the literature.

Model	Mechanism	Scale	Input	Count of catchments	NSE statistics		Reference
					Mean	Median	
LSTM	Data-driven	Local	With static inputs	531	0.52	0.58	This study
PHY-LSTM ¹	Physics-guided	Local	With static inputs	531	0.61	0.66	This study
LSTM	Data-driven	Regional	Without static inputs	531	0.39	0.48	This study
PHY-LSTM	Physics-guided	Regional	Without static inputs	531	0.49	0.56	This study
LSTM	Data-driven	Regional	With static inputs	531	0.69	0.73	This study
PHY-LSTM	Physics-guided	Regional	With static inputs	531	0.67	0.72	This study
XAJ ²	Physical-based	Local	—	531	0.47	0.52	This study
LSTM	Data-driven	Regional	With static inputs	531	—	0.73	Feng et al. (2020)
LSTM	Data-driven	Local	With static inputs	569	0.52	0.60	Jiang et al. (2020)
VIC ³	Physical-based	Local	—	531	0.51	0.55	Newman et al. (2017)
VIC	Physical-based	Regional	—	531	0.17	0.31	Mizukami et al. (2019)
SAC-SMA ⁴	Physical-based	Local	—	241	0.56	0.60	Newman et al. (2017)
mHM ⁵	Physical-based	Local	—	447	0.62	0.66	Mizukami et al. (2019)
mHM	Physical-based	Regional	—	447	0.44	0.53	Mizukami et al. (2019)
HBV ⁶	Physical-based	Local	—	447	0.43	0.54	Seibert et al. (2018)
FUSE ⁷	Physical-based	Local	—	447	0.58	0.62	Kratzert et al. (2019)

1. Physics-guided LSTM. 2. Xinanjiang model (Zhao, 1992); 3. Variable infiltration capacity model; 4. Sacramento soil moisture accounting model; 5. Mesoscale hydrologic model; 6. Hydrologiska Byråns Vattenbalansavdelning model; 7. Framework for understanding structural error model.

3.3. Model evaluations

3.3.1. Simulation accuracy

Four widely used statistical criteria are used to quantify the performance of the model simulations in this study.

(1) Nash-Sutcliffe Efficiency (NSE) (Nash and Sutcliffe, 1970):

$$NSE = 1 - \frac{\sum_{t=1}^{NT} (q_{obs}(t) - q_{sim}(t))^2}{\sum_{t=1}^{NT} (q_{obs}(t) - \bar{q}_{obs})^2} \quad (14)$$

where $q_{obs}(t)$ is the observed variable; $q_{sim}(t)$ is the computed variable; NT is the total number of observations; and $\bar{q}_{obs}$ is the overall mean daily observed variable.

(2) Pearson Correlation Coefficient (r):

$$r = \frac{\sum_{t=1}^{NT} (q_{obs}(t) - \bar{q}_{obs})(q_{sim}(t) - \bar{q}_{sim})}{\sqrt{\sum_{t=1}^{NT} (q_{obs}(t) - \bar{q}_{obs})^2} \sqrt{\sum_{t=1}^{NT} (q_{sim}(t) - \bar{q}_{sim})^2}} \quad (15)$$

(3) Bias (Gupta et al., 1999):

$$Bias = \frac{\sum_{t=1}^{NT} (q_{obs}(t) - q_{sim}(t))}{\sum_{t=1}^{NT} q_{obs}(t)} \times 100\% \quad (16)$$

(4) The Kling-Gupta Efficiency (KGE) (Gupta et al., 2009) is a model evaluation criterion that can be decomposed into the contribution of mean, variance, and correlation on model performance. KGE is calculated as follows:

$$KGE = 1 - \sqrt{(r-1)^2 + (\mu_{sim}/\mu_{obs} - 1)^2 + (\sigma_{sim}/\sigma_{obs} - 1)^2} \quad (17)$$

where μ and σ are the mean and standard deviation of the variables. KGE value ranges from $-\infty$ to 1, with a value closer to 1 indicating a better simulation performance.

3.3.2. Physical consistency

Three criteria are selected to pay attention to the model performance of high discharge and low discharge according to the flow duration curve (FDC) (Yilmaz et al., 2008).

(5) Bias in FDC High-Segment Volume (FHV):

$$FHV = \frac{\sum_{h=1}^H (q_{sim}^{\xi} - q_{obs}^{\xi})}{\sum_{h=1}^H q_{obs}^{\xi}} \times 100\% \quad \xi = 1, 2, \dots, H \quad (18)$$

where H is the number of flood events with a frequency of less than 0.02; and q^{ξ} is the discharge of flood frequency ξ .

(6) Bias in FDC Midsegment Slope (FMS):

$$FMS = \frac{[\log(q_{sim}^{0.2}) - \log(q_{sim}^{0.7})] - [\log(q_{obs}^{0.2}) - \log(q_{obs}^{0.7})]}{[\log(q_{obs}^{0.2}) - \log(q_{obs}^{0.7})]} \times 100\% \quad (19)$$

(7) Bias in FDC Low-Segment Volume (FLV):

$$FLV = -1 \cdot \frac{\sum_{l=1}^L [\log(q_{sim}^l) - \log(q_{obs}^l)] - \sum_{l=1}^L [\log(q_{obs}^l) - \log(q_{obs}^L)]}{\sum_{l=1}^L [\log(q_{obs}^l) - \log(q_{obs}^L)]} \times 100\% \quad (20)$$

$l = 1, 2, \dots, L$

$$q^L = \min[q(t)] \quad (21)$$

where L is the number of low flows with a frequency higher than 0.7. The physical consistency becomes better with the value of three evaluation approaches 0.

4. Results and discussion

The proposed network considering physical mechanisms was implemented into the deep learning model in local and the regional scales, respectively. First, we explained why the input of synthetic samples was used to solve the problem of physical consistency in local models (Section 4.1). Second, we further explained how extreme events of the long-duration heavy rainstorm improved the simulation of high flow (Section 4.2). After that, the prevention of the negative values by extreme events of the long-duration rainless was clarified (Section 4.3) and then the monotonicity of the simulation was verified (Section 4.4). Finally, we analyzed the reasons for the model performance in the regional model (Section 4.5).

4.1. Comparison of model performance of physics-guided deep learning networks in local models

A set of existing LSTM models and some hydrological models by several previous studies are collected over the CAMELS, which were

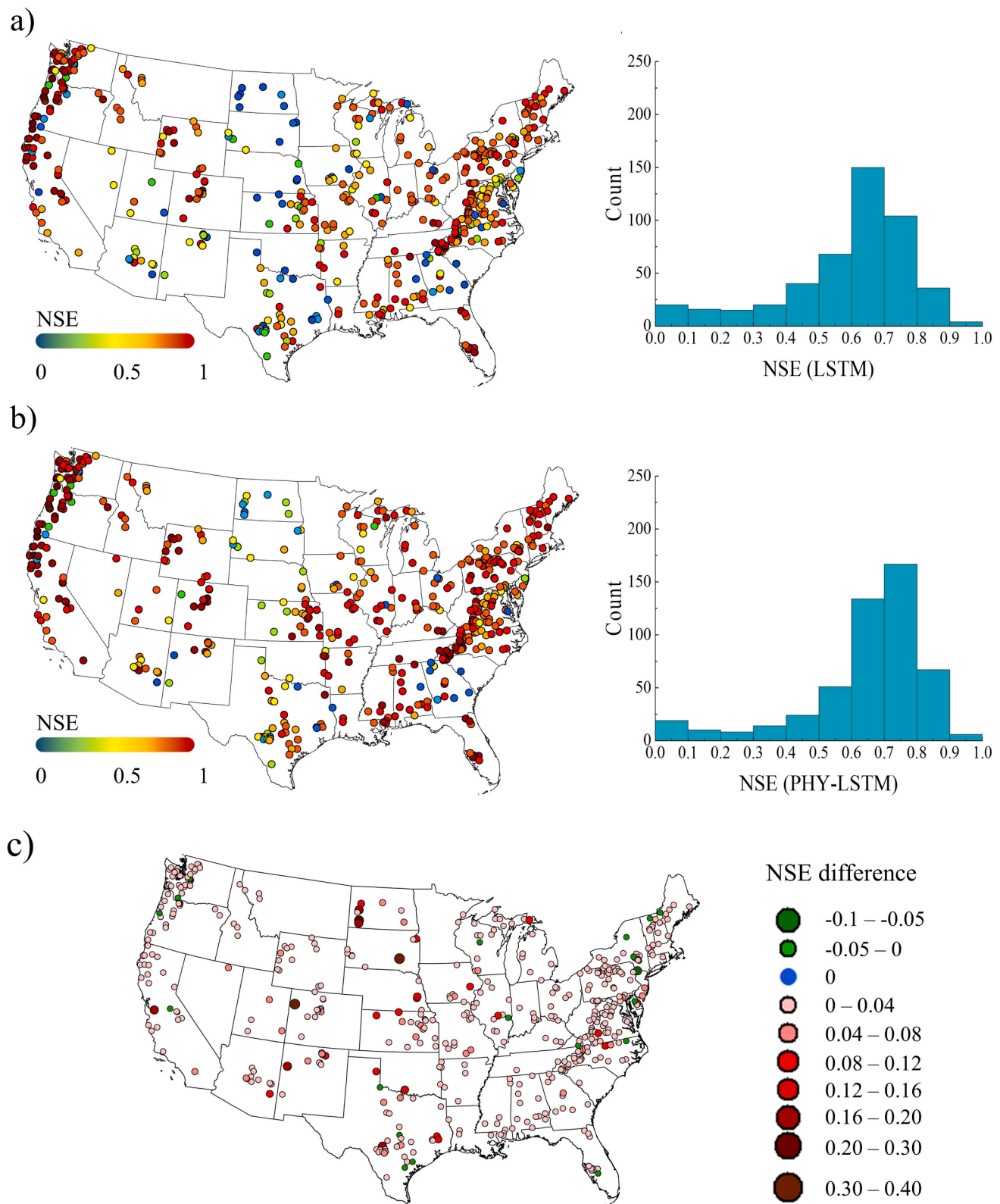


Fig 2. Improvements in performance between (a) conventional LSTM network and (b) PHY-LSTM network in local models during the testing period. c) presents the NSE difference between the LSTM and PHY-LSTM where positive dots mean improvement and negative dots indicate worse PHY-LSTM performance.

calibrated in individual basins or the continental United States. Table 3 shows the results of the models in this study compared with the results of the other models used by the researchers in the CAMELS. The local LSTM models were trained as the base model for comparison from all 531 basins, using inputs of the meteorological data and static catchment

attributes. The results of the base model were the same as the one in Kratzert et al. (2019), which reported a mean NSE of 0.52 in local models. Furthermore, the PHY-LSTM network had a higher simulation accuracy in the testing set. The average NSE was 0.61, the median was 0.66, which was 0.08 and 0.09 higher than the conventional local LSTM

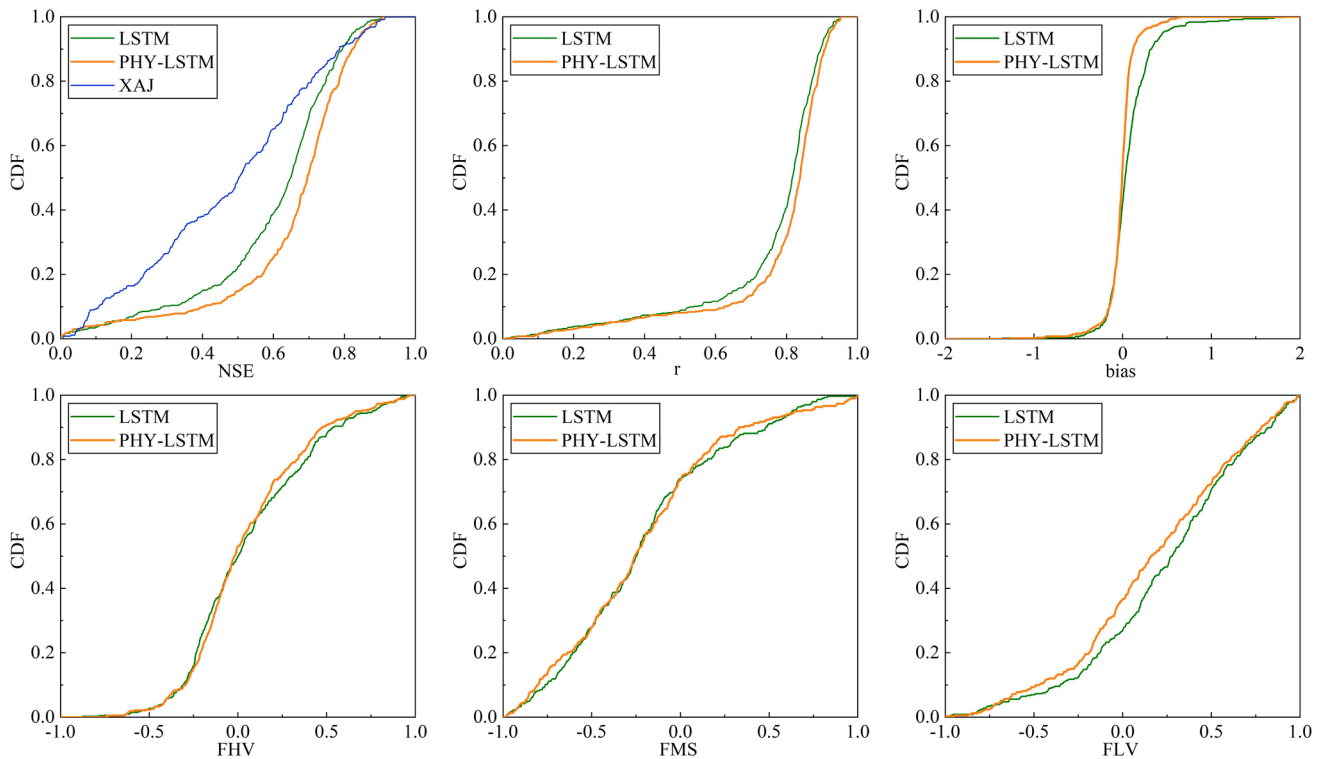


Fig 3. Cumulative density functions of the evaluations between conventional LSTM network and PHY-LSTM network in local models during the testing period.

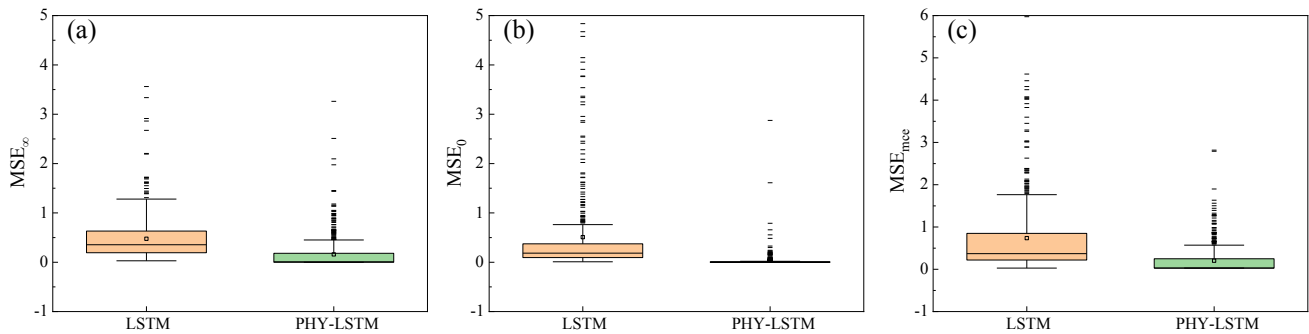


Fig 4. Boxplots of MSE of three physical mechanisms between conventional LSTM network and PHY-LSTM network over the CAMELS during the testing period, a) the improved MSE for heavy rainstorm events, b) the improved MSE in response to the rainless events, c) the improved MSE for the monotonic relationship between rainfall and runoff.

models. The improvement of the PHY-LSTM network lies in the input samples. As a novel input augment method, the synthetic sample are added to improve the PHY-LSTM network considering the physical mechanisms. The results indicated that the input of synthetic samples can be helpful to provide more information on the definite boundaries rarely considered by the actual observations.

As shown in Fig. 2, the conventional LSTM and the PHY-LSTM network were compared with the NSE distribution in the CAMELS during the testing period. The NSE of the conventional model in 26 basins was higher than 80%, 112 basins higher than 70%, and 237 basins higher than 60%. The results of the PHY-LSTM network simulation indicated that the NSE of 52 basins was higher than 80%, that of 190 basins was higher than 70%, and that of 343 basins was higher than 60%. To evaluate the effectiveness of the proposed approaches, the simulation results of the conventional data-driven LSTM model and the physics-based Xinanjiang model are compared in the same basins. Fig. 3 shows the cumulative probability distribution of the six metrics of PHY-LSTM networks in all studied basins. There was a significant difference

between LSTMs trained with vs. without physical mechanisms. The mean r improved from 0.78 to 0.89, while simultaneously reducing the bias. Then three criteria are compared according to the flood frequency curve. FLV and FHV improved, but there was no statistically significant difference for FMS. Overall, the results showed that the proposed PHY-LSTM network can simulate the hydrological processes effectively.

4.2. High flow simulation by deep learning about extreme events of the long-duration heavy rainstorm

To overcome the problem of inadequate mechanisms for extreme events, synthetic rainstorm samples assisted in training physics-guided deep learning models. As shown in Fig. 4a, the MSE of the heavy rainstorm was reduced using the deep learning model based on a physical mechanism compared with the conventional model. The MSE of the heavy rainstorm constraints decreased from a mean of 0.47 to a mean of 0.15. The performance of the conventional LSTM model in high flow simulation was poor because of the rarity of rainstorm samples or the

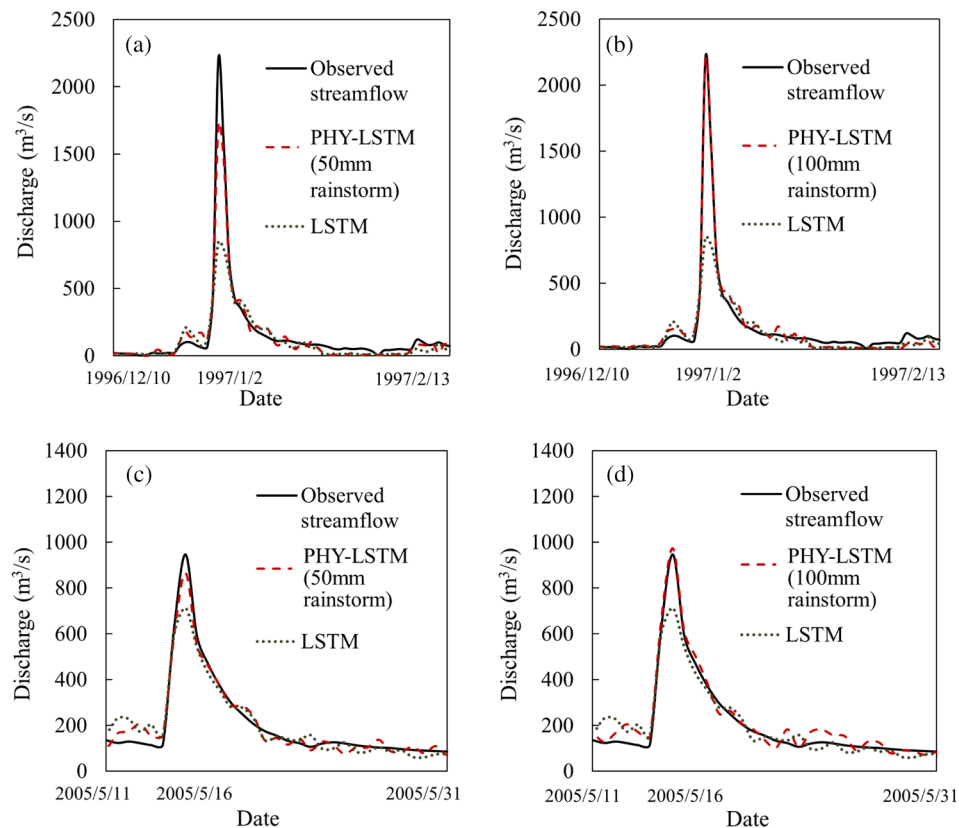


Fig 5. Improved flood peak simulation by extreme events of the long-duration heavy rainstorm in a typically selected basin (i.e., Pitman River, CA, basin number: 11237500), and different intensities of heavy rainfall samples have different improvements on flood peaks.

absence of heavy rainstorm events in the training period. The simulated peak flow was often unsatisfactory, especially in the case of heavy rainstorm events with low frequency. Therefore, the rainstorm samples were added to the historical observations for a long time. A typical basin (i.e., Pitman River, CA, basin number: 11237500) is selected as an example. As shown in Fig. 5, the flood peak flow simulated by the model was greatly improved compared with the conventional LSTM simulation, when adding a 50 mm long-duration rainstorm and 100 mm long-duration rainstorm respectively in the selected basin. At this moment, all the rainfall should generate runoff when the soil moisture was saturated after a long period of the rainstorm. The results demonstrated that the synthetic samples of a heavy rainstorm can effectively improve flood peak simulation, however, it was related to the intensity of the rainstorm. The rainstorm of 100 mm was the better intensity of the rainstorm flood in the selected basin. The results confirmed that the intensity of the samples was associated with the improvement.

4.3. Low flow simulation by deep learning about extreme events of the long-duration rainless

To improve the ability of deep learning networks to simulate low flows, synthetic samples of long-duration rainless events were used. As shown in Fig. 4b, the MSE of the long-duration rainless constraint decreased from mean 0.50 to mean 0.04. If there was no rain for a long time, the soil moisture would be exhausted and runoff would not happen again. However, as shown in Fig. 6a, the conventional simulation often appeared negative flow, which indicated that the existing samples cannot train the model in the part of low flow. The usual experience was to change the negative value to 0 directly by the ReLU method. In this study, the conditions of synthetic rainless samples can be simulated confidently to constrain the conventional model. In Fig. 6b, the occurrence of negative flow was greatly prevented with the support of

constraints of long-duration rainless events. The addition of synthetic samples was helpful to improve physical consistency, which provided proof for the establishment of the data-driven model using theoretical constraints.

4.4. Extending the ability of the monotonic rainfall-runoff relationship

The monotonic relationship between rain (or evaporation) and runoff was used to train the PHY-LSTM network. The monotonicity rule says that hydrological simulation should have monotonically output with monotonical inputs, that is, the runoff would increase even if a small increase in the rainfall. As shown in Fig. 7a, a small disturbance is increased in the rainfall to the testing set of the LSTM model, and the results showed that there was a huge nonmonotonicity in most basins. The results of most basins showed that the conventional LSTM model has high MSE. The mean and median MSE_{mcp} of the 531 basins showed that the monotonic performance of the conventional deep learning network was gradually getting worse during the learning epochs. The conventional data-driven model mainly depended on the information contained in the data, which was applicable in areas equipped with a large supply of representative data samples. As shown in Fig. 4c, the MSE of the monotonicity decreased from mean 0.36 to mean 0.11. For a small increase in the historical rainfall samples, the new simulation results should increase monotonously compared with the previous simulations. Fig. 7b illustrated the decreased MSE by the PHY-LSTM network, where the monotonic relationship educated the deep learning models. It provided the largest set of significant clusters of the approach and the possibility of a deep learning model to extend the ability of monotonicity. PHY-LSTM models can be well suited to represent understandable processes using known scientific principles.

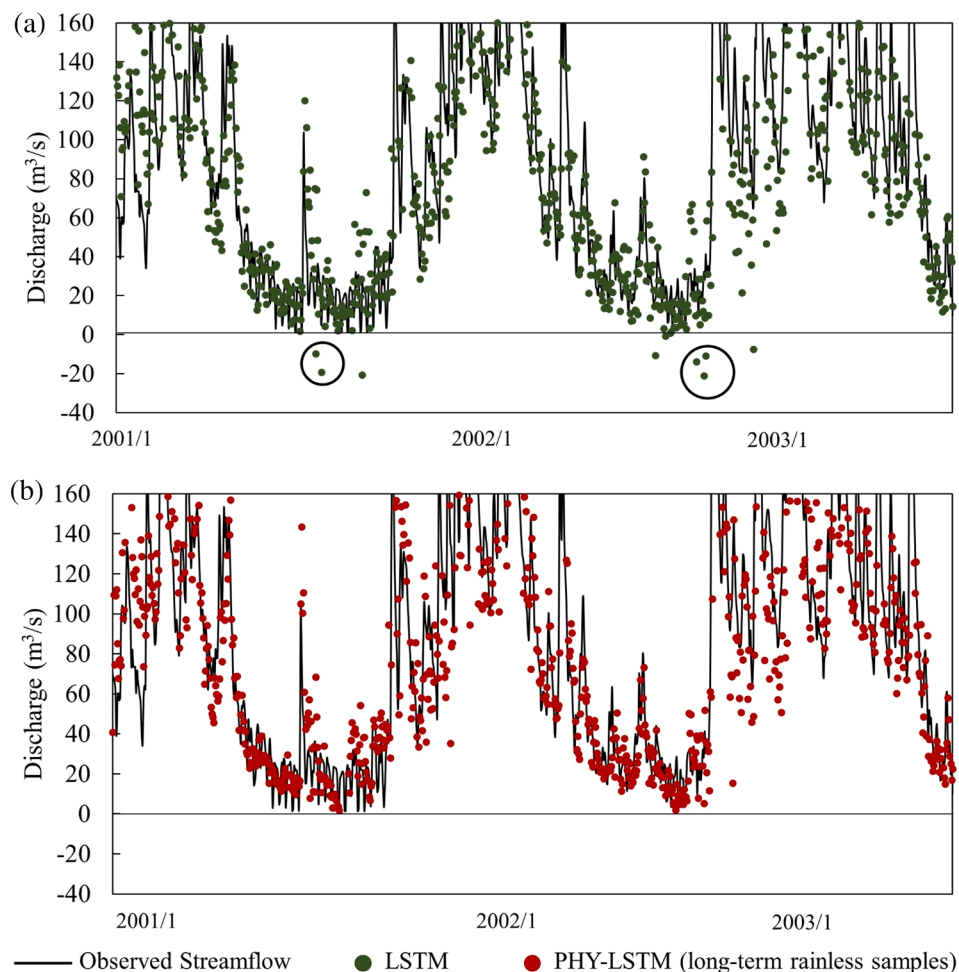


Fig. 6. Low flow simulation with the extreme events of long-duration rainless in a typically selected basin using (a) conventional LSTM network and (b) PHY-LSTM network.

4.5. Model performance of physics-guided deep learning networks in the regional model

As shown in Table 3, in the regional model, the PHY-LSTM model improved the simulation accuracy without catchment attributes added to the input data, when the mean NSE value increased from 0.39 to 0.49. However, another important finding was that the PHY-LSTM was not improved with catchment attributes in the regional model. Table 4 shows all evaluations for the local and regional models. To find out why physical constraints were effective for local models but had little improvement in the regional model, a set of experiments were performed to compare the simulation accuracy of LSTM and PHY-LSTM with the increasing number of basins. As shown in Fig. 8, the deep learning network was less effective when the number of basins was small or even in local models, and samples of hydro-physical processes were used as effective information to improve the simulation accuracy. The accuracy of the deep learning network improved as the increasing number of basins when more basins can provide more information to each other. The diminishing role played by physical constraints suggested that the regional deep learning model was not entirely non-physical. A possible explanation for these results may be extreme weather events that did not occur frequently in one basin but other basins of the region. The model can mine the rainfall-runoff relationship among basins with similar characteristics through the deep learning of the regional LSTM model. Sufficient data not only improved the accuracy of the model but also contributed to the physical consistency of the model. The further comparison indicates better FHV and FLV for the

regional model. However, Fig. 9 shows the monotonic performance of the change of the discharge with an increase of 0.1 mm rainfall disturbance, and the results demonstrated that the nonmonotonic rainfall-runoff response still existed even in the regional model. Together these results provide important insights into the monotonic relationship, which should be considered by physics-guided deep learning network although in the regional model.

5. Conclusions

In this paper, a deep learning model based on physical mechanisms was constructed to simulate the rainfall-runoff process over the CAMELS in the United States. The main objective of the current study was to investigate the effects of extreme events and the monotonic relationship of a deep learning network for rainfall-runoff simulation. Considering the physical mechanisms, synthetic samples were used as extra inputs for deep learning, and the exterior loss functions of the hydrological physical process were added to the conventional optimization objective of the model. Conclusions derived are as follows:

- (1) It was indicated that synthetic samples provided more information on the hydrological theory that is not embedded in the actual observations. The mean NSE increased from 0.52 for models without physical consistency to 0.61 for local models based on physical mechanisms during the testing period.
- (2) Extreme events were effective for establishing deep learning models with physical consistency. Adding synthetic samples can

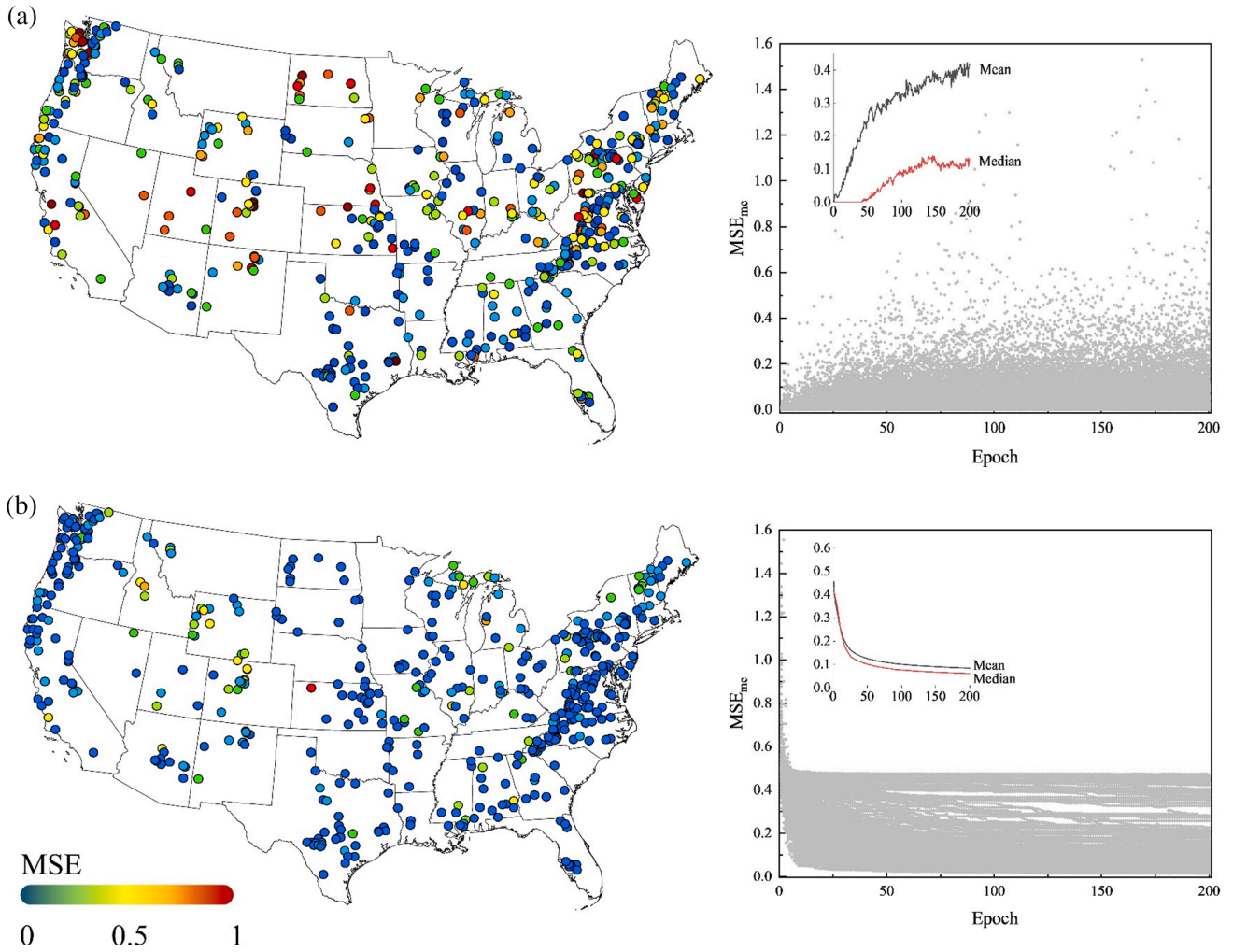


Fig 7. Spatial distributions and epoch trends of MSE for the monotonicity performance between (a) conventional LSTM network and (b) PHY-LSTM network over the CAMELS during the testing period.

Table 4

Comparison of model evaluations for the testing dataset in four different experiments.

Model		Local model		Regional model	
		LSTM	PHY-LSTM	LSTM	PHY-LSTM
NSE	Mean	0.52	0.61	0.69	0.67
	Median	0.58	0.66	0.73	0.72
KGE	Mean	0.49	0.59	0.69	0.68
	Median	0.55	0.62	0.75	0.75
r	Mean	0.78	0.80	0.89	0.89
	Median	0.83	0.86	0.94	0.93
Bias	Mean	-0.07	-0.01	-0.01	-0.01
	Median	-0.07	-0.02	-0.01	-0.01
FHV (%)	Mean	-25.30	-13.74	-18.06	-19.66
	Median	-24.32	-20.82	-14.80	-16.21
FMS (%)	Mean	-82.16	-35.21	-11.69	-10.87
	Median	-15.57	-14.33	-8.30	-8.15
FLV (%)	Mean	26.10	24.29	25.11	27.52
	Median	32.48	28.51	26.50	30.40

improve the simulation of flood peaks with the MSE of the heavy rainstorm constraints decreased from mean 0.47 to mean 0.15. Besides, synthetic samples can reduce the occurrence of negative streamflow with the MSE of long-duration rainless constraints decreased from mean 0.50 to mean 0.04. This approach is useful

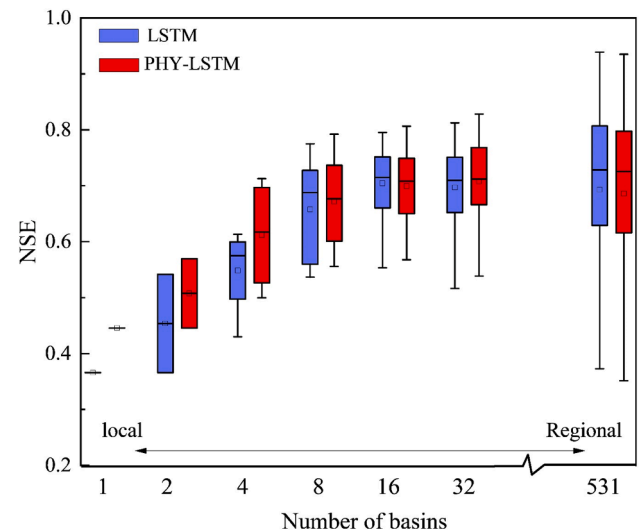


Fig 8. Comparison of the simulation accuracy between LSTM and PHY-LSTM with the increasing number of basins from local to regional models.

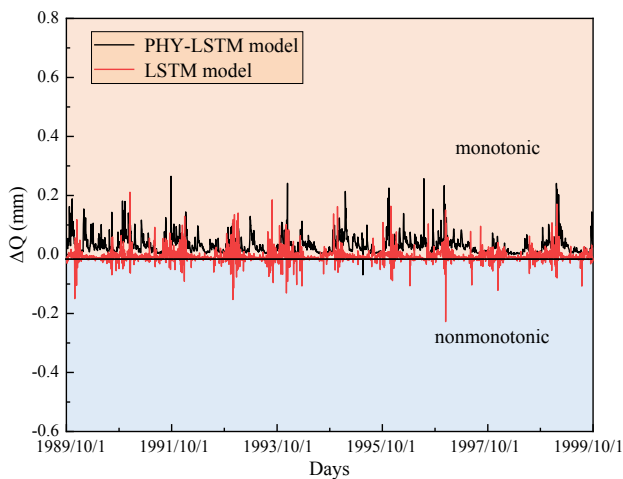


Fig 9. Change in the runoff with an increase of 0.1 mm rainfall disturbance in a typically selected basin, where positive values are monotonic and negative values are non-monotonic.

for hydrological simulation, especially for the prediction of flood peaks and the prevention of negative values in the simulation.

- (3) The relationship between rainfall and runoff in this study can train deep learning networks both in local and regional models, which can significantly extend the ability of the monotonicity. The deep learning network based on constraints of physical consistency enables the system to train the model about available physical knowledge.

Further research should focus on validating the effects of more constraints of physical consistency and expanding the use of the physics-guided deep learning model in the field of water resources.

CRediT authorship contribution statement

Kang Xie: Conceptualization, Methodology, Writing – original draft.
Pan Liu: Conceptualization, Supervision, Writing – review & editing.
Jianyun Zhang: Supervision, Writing – review & editing. **Dongyang Han:** . **Guoqing Wang:** Writing – review & editing. **Chaopeng Shen:** Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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